2023 New Zealand (Aotearoa) Biosciences Expedition Proposal

Assessment of Microbial Diversity Based on Soil Physiochemistry

PROPOSED BY

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1. Executive Summary

Mitigation of the impact of climate change on biodiversity requires knowledge of local microbial community composition. Although unseen, above-ground biodiversity is influenced by the composition of subsoil microorganisms, hence soil microbial composition is critical in formulating and monitoring conservation methods for soil health and habitat biodiversity. However, the inclusion of soil microbiology in land management is often used belatedly or disregarded (Winder, 2003), and guidance on quick, accurate measurements isn't well established (Fierer et al., 2021) or affordable.

Within terrestrial environments, soil microbiome compositions are highly reflective of soil health, and underpin supporting ecosystem services. Preservation of these ecosystem services within agriculture (FAO, 2016) is crucial to sustainable food development (Sustainable Development Goals 2,12) as population growth continues to increase. Local biodiversity is also important to retaining soil resilience to the climate change impacts on land productivity and may be used in monitoring and managing agricultural systems' vulnerability to sustained damage following ecological disturbances.

'Regenerative agriculture' (RA) generally describes land management efforts to preserve soil function and biodiversity, while still sustaining agricultural productivity and associated economic value (Hermans et al., 2020). As microbial biodiversity influences soil functionality, assessing the impact of RA across study sites on soil health requires the investigation of microbial community composition as guiding evidence for land management practices. We aim to characterise the soil microbiome of various microhabitats, which has only been done in microhabitats managed with RA to a limited extent (Hermans et al., 2023).

Microbial composition will be directly measured (qPCR) as well as inferred through microhabitat physiochemistry. Soil physiochemistry is a "dominant determinant" (Louisson *et al.*, 2023) of microbial community organisation, so profiling soil quality across heterogenous study sites will allow us to estimate microorganism biodiversity and local habitat resiliency. Physiochemical features will also validate which features are more accurate assessors. These features can also be assessed as microbial biodiversity indicators to suggest low-cost or field-testing methods that are operable by farmers or operations lacking funds or access for laboratory assays.

Further understanding of how soil features across microhabitats reflect local community biodiversity will allow better monitoring of microhabitat health and resilience to ecological ecosystem function monitoring in the face of ecological disturbances or changes in land use.

Managing biodiversity across varying habitats involves the integration of scientific research with local community values. Scientific and policy ecological preservation efforts involve significant communal and cultural interaction, particularly with indigenous peoples; 85% of globally protected areas for biodiversity are inhabited by indigenous peoples (Alcorn, 2000). Historical disregard of indigenous knowledge systems (amongst the scientific community) (DeWalt, 1994) has led to the disconnection of the scientific community from local communities offering interdisciplinary, observation-based, locally-conscientious, and low-disturbance (Toledo, 2001) methods of resource utilisation. Indigenous agricultural knowledge has been characterised by “diverse, adaptive strategies” (DeWalt, 1994), which comprise low-disturbance conservation methods.

This expedition offers the invaluable integration of ecological sciences and mātauranga Māori (indigenous knowledge) by partnering with Ngāti Whātua Ōrākei (Māori community) and The Living Laboratories Project, currently developing RA methods to promote indigenous and old-growth forests.

In a unique, collaborative research exploration with the Auckland University of Technology (AUT), we will assess soil physiochemistry and microbial diversity across indigenous sampling sites under varying successional stages and management plans. Measurements will assess physiochemical indicators of microbial diversity, and characterise the microbial communities of (RA) soil microhabitats.

2. Expedition Introduction

2.1 Brief Aims and Objectives

Our project aims to identify how physicochemical microbiomes differ between microhabitats with varying degrees of biodiversity – including that of native New Zealand species that The Living Labs are aiming to restore nationwide. Our results will aid those efforts by providing evidence of the effect of the succession stages and biodiversity of these native species on microbe biodiversity. It will also provide evidence of the efficacy of low-cost direct and indirect measurements of microbe biodiversity.

Through the month of August, our expedition group will profile soil physiochemistry to characterise soil microbial biodiversity across microhabitats, confirming microorganism community composition with qPCR molecular analysis. As a partner of the Living Laboratories, Auckland University of Technology (AUT) has offered its resources and facilities, including samples collected from when regenerative efforts first began at the sites, to assist our expedition soil sampling and testing, as well as connecting us with testing (material) suppliers

We will analyse microbial diversity across sample sites varying in land management and successional stages (including native and old-growth plant composition). Heterogeneous site physiochemistry will be used to estimate soil quality, estimate soil microbial communities, assess local resiliency against environmental disturbances, and provide evidence for low-cost measurements of microbiome biodiversity.

2.2 Dates

At the end of July, the expedition team will travel to Auckland, New Zealand for a period of four weeks before returning at the end of August. Sampling will begin on the 2nd August and the sampling and testing will be conducted in triplicate, with sampling and analytical procedures each taking four days. In the last week of the expedition, time will be taken to review the sampled data and conduct further tests if necessary.

The week after returning to the UK will be used to finalise the preliminary project report, and the following month to synthesise the final report.

2.3 Location

The site of study is a 2.2-hectare field site within the Pourewa Creek Recreation Reserve, Ōrākei, and contains a variety of site preparation, nursery treatments, and plant growth experiments, which will be used to produce soil samples of different successional stages and microhabitats. The field site is located on the southern edge of Takaparawhā (Bastion Point) in Tāmaki Makaurau (Auckland).

This part of the Pourewa estuary is co-managed by the AUT Living Laboratories Project and Ngāti Whātua Ōrākei Whai Māia. Maps below are of (*Fig.1*) the Ōrākei community, including the Pourewa Creek Recreation reserve where field testing sites are located, and the larger Auckland area where field sites, laboratory facilities, and accommodation will be within (*Fig.2*).

Sample sites are accessible through public transportation, vehicle sharing with citizen scientists, Living Laboratories members, or AUT vehicles.

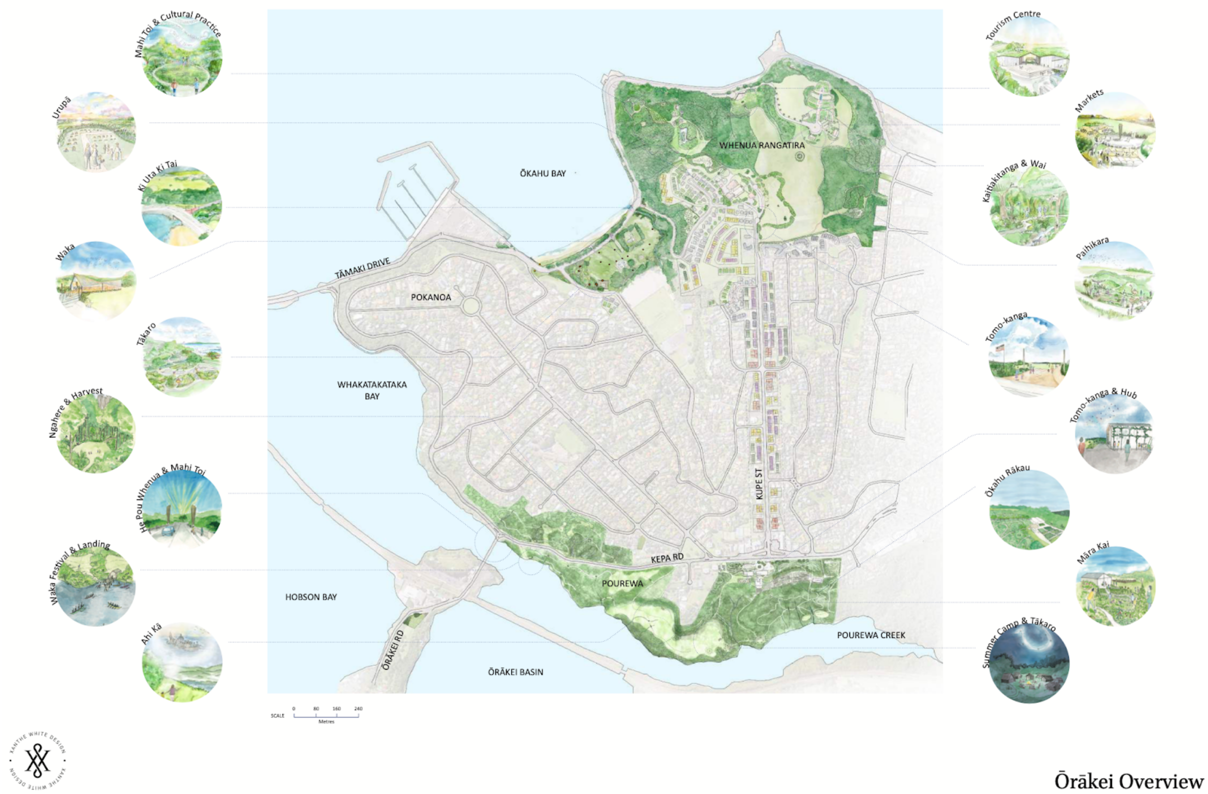


Figure 1. *The Ōrākei community, including the Pourewa Creek Recreation Reserve, boarding the Ōrākei Bay on the southside of Bastion Point.*

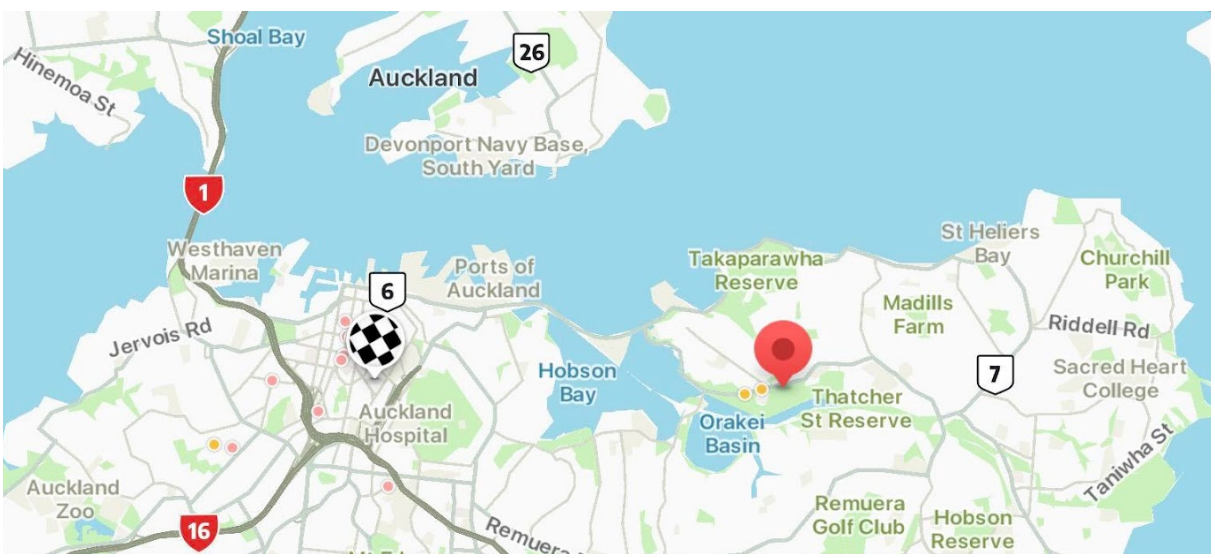


Figure 2. *Within the larger Auckland area, pins mark the Ōrākei community (red pin), where sampling will be conducted, relative to the City Campus of AUT (checked pin), where accommodation and testing facilities will be near.*

3. Expedition Aims and Methodology

Aims

- Identification of how physiochemical microbiomes differ between microhabitats with varying degrees of biodiversity.
- Provide evidence on the effect of succession stages and biodiversity of native species on microbe biodiversity.
- Evidence of the efficacy of low-cost, direct and indirect microbe biodiversity measurements.

Objectives

3.1. Estimate soil microbial diversity and community composition through soil physiology.

3.1.1 Soil Texture is well considered in valuing soil water retention, nutrient storage, and nutrient cycling capacity. While soil heterogeneity underpins microhabitat diversity (Curd et al., 2018), its relationship to microbial composition, another integral factor in biodiversity, has not been well characterised (Xia et al., 2020). In an indirect assessment of soil microbial diversity, soil texture can be affordably and quickly characterised through **hydrometer analysis** (Hossain et al., 2021) with much greater accuracy than estimations in the field could achieve. Clay, sand, and silt composition also relates to mineralized nutrient content and material retention, [indicating] the need for multifactorial analysis related to bacterial diversity.

3.1.2 Soil pH is critically valued in soil quality assessments and is the ultimate predictor of microbial local (α) diversity changes driven by anthropogenic global change factors (Zhou et al., 2020) such as climate change or land-use change. pH testing may be done without the need for high-throughput laboratory methods through **pH test strips and a colorimeter app downloaded on a portable device**.

3.1.3 Soil (Gravimetric) Water Content and Bulk Density affect microbial diversity on a microhabitat, temporally-variable scale as water is a fundamental abiotic factor, affecting fundamental microorganism niches directly and indirectly (as a carrier for leached soil nutrients and determinant of plant community composition). Water content and bulk density may be measured through **oven drying** and **massing** as detailed in Xia et al. (2020). ***Water holding capacity*** may additionally be measured by **massing after 24h of water saturation** (Zhang et al., 2011).

3.2. Estimate soil microbial diversity and community composition through soil nutrient (levels).

3.2.1 Active Carbon (C) Content is the "currency of nature" (Breker, 2021) and underpins the most fundamental biological processes. C, amongst the following nutrients, is involved in competitive soil interactions which drive microbial biodiversity; nutrient heterogeneity shapes niche and species diversity (Torsvik & Øvreås, 2002). **Active C** (permanganate-oxidizable carbon (POXC); $KMnO_4$) measurement (*Active Carbon (POXC)*, *2020*; Weil et al., 2009) accurately and reliably measures Soil Organic Matter i.e. biologically-active C which readily oxidises. We will follow the method described in *Active Carbon (POXC)* (2020).

For sample testing, 1 L solution of 0.015 M $KMnO_4$ and 0.1 M $CaCl_2$ in water is necessary for testing of all 45 samples, equating to 2.38 g and 14.70 g of $KMnO_4$ and $CaCl_2$ respectively. Colour change will be measured with a spectrophotometer or colourimeter, potentially available at AUT, and/or with a colourimeter app.

3.3. Genetic Measures of Soil Microorganisms through qPCR.

3.3.1 Quantitative PCR (qPCR) analysis would quantify microbial presence across soil samples through the real-time measurement of present genetic material through a combination of PCR amplification reactions and DNA-intercalating probes (Sigma-Alrich) for real-time product detection. qPCR is a standard molecular procedure well-characterised to assess microorganism community composition (Classen et al., 2010; Fierer et al, 2005) across sample sites.

3.4. Sampling.

3.4.1 Microhabitat Sites variety is key in our analysis to correctly estimate the influence of it on microbiomes. Microhabitats will vary in successional stages, nursery species, and regeneration and biodiversity management practices. We will use 5 microhabitat sites within Pourewa Creek Recreation Reserve.

3.4.2 Repeat visits are necessary to conduct triplicate experiments to gain an accurate picture of physiochemical properties regardless of recent weather and changes in anthropogenic activity. We will visit the site on 3 different days, across 3 weeks, shown on the Gantt chart, and collect repeats of the data, with every site's data being collected on the same day. Time is scheduled in the final week to conduct other repeats if necessary.

4. Preliminary Financial Estimates

Task	Items	Function/Purpose	Cost Estimate (£GBP/ $NZD)
Travel			
Travel documents	New Zealand Electronic Travel Authority and International Visitor Conservation and Tourism Levy	Legal entry into NZ	£17.20/$35 x 4
Flights	Return flight (London to Auckland, *27/07 - 18/08)*	To get to Auckland	£1,195 - 1,900 x 4 *Total estimate uses 1,195 GBP price, which is contingent upon booking time*
Airport Transportation	Airport Taxi	Transport from Airport to Accommodation and back	£44.29/$45 x 2
Accommodation	Airbnb accommodation from approx. 29/07 to 18/08	Housing for the duration of the expedition for all (4) members	£1656 *Living costs for food, and utilities will be personally financed.*
Sample site transportation	Transportation should be available through ride-sharing, but public transport routes are available. Uber is also possible.	Reaching field sites to collect samples	£86.61/$44 *AUT has informally offered transportation free of charge.*
Sampling and Analysis			
3.1.1-3.1.2 Texture, pH	Multi-soil feature tester	Soil pH, temperature, and moisture measurements	£22.09/$44.90
3.1.2 pH	Colorimeter app, pH test strips	Quantitative, colour-based test measurement	£4/$4
3.2.1 Active C content	Potassium permanganate, calcium chloride, distilled water.	POCX test for C measurement	£0/$0 *Small amounts of reagents required. AUT has informally offered to help access them.*
3.3.1 qPCR	Soil sample cost	Direct microbial diversity measurement	£4.61/sample x 45 (~£207) *qPCR ground costs will be covered by the generous collaboration of AUT, and the expedition group will be*

			responsible for sample-by-sample costs only. *Durham laboratory estimates lower costs, at £2-3 per sample which equates to £90-135 total. qPCR costs in New Zealand are likely to be slightly higher than in the UK, but may be less than our estimate.*
Intended Personal Contributions			
Esme Padgett			£650
Ben Keijzer			£650
Hugh Halsey			£650
Anna Daniel			£650
Total (minus Personal Contributions)			**£4,314.73**
Total + 15% Contingency			**£4,961.94**

5. Expedition Member Details

Esme Padgett

2nd year, Biosciences (MBiol), St. Chad's College.

During the summer of 2022, I worked in the Insect Neuroscience Laboratory conducting molecular ecology research on olfactory preferences for flower volatile organic compounds (VOC) in bees (*Bombus spp.*). I conducted field sample collections, *Bombus* and plant species identification through morphological and genotyping methods, and GCMS VOC analysis. I became well-practised in minimally invasive field sampling and standard molecular biology procedures like DNA extraction and imaging, PCR, and gel electrophoresis. It also taught me how to work collaboratively in an interdisciplinary scientific setting, and reinforced my passion for the intersection of molecular biology and ecology.

I also have over a year of combined field sampling experience in estuary and marine systems of the Chesapeake Bay watershed, monitoring water turbidity and E. coli levels and creating a nursery for endangered, local aquatic vegetation with local conservation organisations.

I've served as one of the two International Representatives at my college this academic year, coordinating speakers for a seminar series on culture, heritage, and diversity. This has helped me build avenues of connection-focused communication with space for disagreement and collaboration, and manage group discussions to emphasise open-minded, culturally-sensitive learning.

Additionally, serving as a sports captain and vice-captain at the high school and collegiate levels demonstrates my strong time management skills to pace work efficiently and manage shared goals. My communication style helps me to delegate tasks amongst teammates while maintaining a growth-oriented environment.

Anna Daniel

2nd year, Natural Sciences (Biosciences and Chemistry), St. Chad's College.

In high school, I completed a project on wildfire impacts on soil chemistry, which I presented at a regional science fair and received a bronze certificate for (so I know my way around a soil sampling kit!). Additionally, I spent two weeks conducting wildlife surveys on a community-owned game reserve, gaining an understanding of how research in collaboration with indigenous communities feeds into a larger need for decolonisation.

As a South African, I have a personal interest in learning how initiatives like the Living Laboratories project can be a model for other countries grappling with the dual problems of colonial histories and ecological change due to climate change and human impact. As a natural sciences student, I am familiar with chemistry laboratory skills relevant to testing active carbon. I have strong data analysis skills and motivation, shown by my previous work as a Students' Events Coordinator and a computational biology placement in statistical genomics for the coming year. Finally, my tenure as vice-president of the university's International Students' Association demonstrates strong collaborative and leadership skills which I am excited to apply to an expedition.

Ben Keijzer

2nd year, Biosciences (BSc), Josephine Butler College.

In my time at the University of Durham I have gained experience in ecological data collection methods, including plant identification and various nocturnal monitoring methods (camera/small rodent trapping etc.) and collating and analysing field data into succinct and coherent reports. Further, I am proficient in the programming software R studio, useful for the analyses and display of our results. I have experience organising and working as a group through this and my previous jobs in catering and fast food. This has developed my leadership and teamwork skills through being in charge of the front of house section, as well as my communication and relationship forming ability, through customer interactions. My degree specialisation within biomedicine has provided me with microbiology experience, identifying bacterial species, using colony morphology eye tests and biochemical tests including Gram-staining and API strips, used to identify patient bacterial infections. This developed my knowledge of greater microbial function in the context of gut microbiota, host pathogenesis and immune evasion. Alongside understanding how they function, utilise nutrients and metal ligands, although in the context of human invasion, these skills are adaptable to soil microbes.

Hugh Halsey

1st year, Biosciences (MBiol), St. Chad's College.

From February to August 2022, I worked with the conservation charity A Rocha in Vancouver, Canada. I worked on a wide range of fieldwork projects including water quality surveying, invasive species removal, and endangered species preservation for the Western Toad and Oregon Forestsnail amongst others. I designed surveying strategies to estimate snail population sizes for the Oregon Forestsnail in the Little Campbell Watershed before writing a full report about the status of the species for the charity. The charity worked closely with the Semiahmoo First Nations which gave me a newfound appreciation for the important role indigenous knowledge plays in conservation.
I have also worked in a research laboratory where I supported the work of postgraduate students exploring Duckweed (*Lemna*) in different nitrogen concentrations and temperatures to explore the impact on growth. I gained experience in genetic analysis, carrying out PCR and gel electrophoresis.
I also have previous experience in soil sampling when I conducted my Geography research project at A level. I explored the soil water content of a woodland and field before writing a full report. I received an A* for my Geography A level.

6. Preliminary Travel Details

Visa Requirements

All expedition members qualify for visas on arrival. $52 NZD (£25.60 GBP) each is required for a New Zealand Electronic Travel Authority and International Visitor Conservation and Tourism Levy to be paid in advance. Processing time is typically 72 hours.

Travel

Preliminary flight dates from London to Auckland and return are the 29th of July and 28th of August, respectively. Flexibility for flight dates based on ticket costs.

Accommodation

Accommodation will be within Auckland City Center between to the Central AUT campus and Pourewa. Accommodation in a short-let flat via AirBnb etc is the most cost-effective option.

7. Safety and Medical Details

All expedition members are fully immunised including for COVID-19 and are in general good physical health. New Zealand has a reciprocal health agreement with the UK for UK nationals on short-term visits. Non-UK nationals in the group have travel health insurance and UK nationals will secure appropriate health insurance if necessary individually.

The field site and laboratory are situated within Auckland, a major city with easy access to high-quality healthcare facilities. All expedition members have laboratory experience, whether through coursework or otherwise, and AUT will ensure we have supervision if necessary.

8. Permissions and Arrangements

Field site sampling permissions have been given by Dr. Syrie Hermans on behalf of The Living Laboratories Project, which shall provide permits to conduct soil field surveys and sampling. Access to AUT facilities and equipment is generously provided by Dr. Syrie Hermans. The expedition will have to reimburse the sample-by-sample costs (e.g. qPCR), but AUT has offered to cover some testing costs for analyses relevant to their existing research interests. Therefore Sampling and Analyse costs (see 4. Preliminary Financial Estimates) are likely to be reduced thanks to the generous collaboration with AUT.

Pourewa is Ngāti Whātua Ōrākei (Māori)-owned, and is involved with ecological science through the Living Laboratories. The Ngāti Whātua Ōrākei Whai Māia generously offers Pourewa land for responsible cultural, social, and environmental aspirations, including community learning and scientific study. It is with the deepest gratitude and respect that we will conduct ecological tests on the Pourewa Creek Recreation Reserve.

9. Advisor Contacts

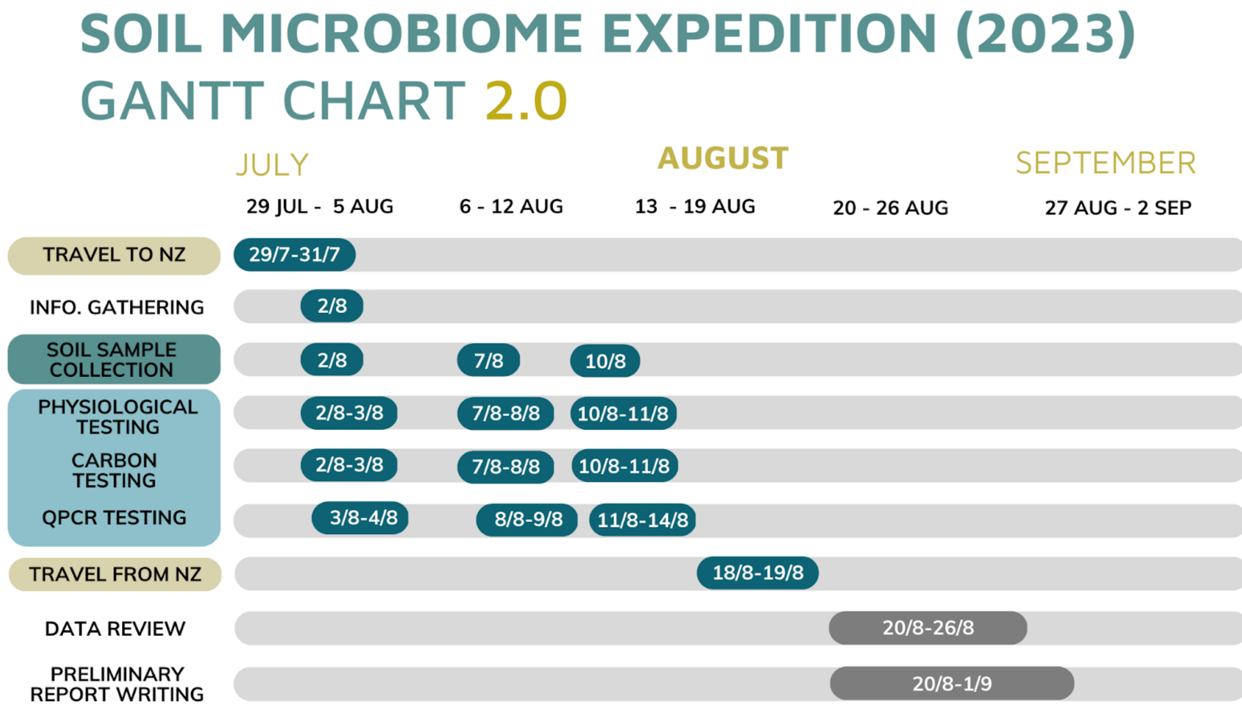
Expedition Group Contact:

Robert Baxter (robert.baxter@durham.ac.uk)

The Living Laboratories Project, with AUT:

Syrie Hermans (syrie.hermans@aut.ac.nz)

10. Timeline



11. Risk Assessment

Full Name of Student Group	*Durham University Expeditions Group 2023*
Description of task or Activity:	*Expedition to New Zealand, visiting sites, collecting and analysing soil samples for its resident microbiota*
Location(s), Date, Time:	*Pourewa Creek Recreation Reserve, Ōrākei* *29th of July – 18th of August*

Hazards	Those at risk	How could they be harmed?	Uncontrolled risk level	Required controls	Controlled risk level
Covid 19 transmission due to overcrowding	Participants, Staff, Other Members of the Public	Infected with Covid 19 with the potential for breathing difficulties, hospitalisation, death	**5X4 High Risk**	The safe Covid capacity of the space will be identified and adhered to (this includes participants, coaches and instructors). Masks are advised but not mandatory. We will adhere to the guidelines given by local areas and external venues.	**2X2 Low Risk**
Covid 19 transmission by contact with an infected individual	Participants, Staff, Other Members of the Public	Infected with Covid 19 with the potential for breathing difficulties, hospitalization, death	**5X4 High Risk**	If we show symptoms we must not attend the expedition or remain isolated if symptoms onset during the trip.	**2X2 Low Risk**
Covid 19 transmission due to social distancing being flouted	Participants, Staff, Other Members of the Public	Infected with Covid 19 with the potential for breathing difficulties, hospitalisation, death	**5X4 High Risk**	Respect other people's space wherever possible, seek consent	**2X2 Low Risk**

Covid 19 transmission from sharing items and equipment	Participants, Staff, Other Members of the Public	Infected with Covid 19 with the potential for breathing difficulties, hospitalisation, death	**5X4 High Risk**	Participants will bring their own items where possible. Where this is not reasonably practicable and the sharing of equipment is essential to the expedition, equipment will be cleaned with sanitising wipes. Participants will also be asked to use hand sanitiser frequently. Food and drink can be shared, we will sanitizing before and after handling items.	**2X2 Low Risk**
Travel to and from the locations	Participants	Become lost, miss transport pick up/drop off. Crash, delays, cancellations.	**3X3 Medium Risk**	Public transport will be used to the sites. Pick up meeting place and times will be communicated to participants well in advance. Extra time will be allowed in case of traffic/delays. The rules of the road will be adhered to at all times.	**3X1 Low Risk**
Bad weather	Participants	Cold, damp, wet, too hot	**4X2 Medium Risk**	The weather will be checked in advance and also on the day of the trip/excursion. We will wear appropriate clothing. In cases of serious bad weather the expedition may be cancelled.	**4X1 Low Risk**
Participants becoming lost	Participants	Become lost/ disorientated/ separated from group, physical/ mental injury	**3X3 Medium Risk**	A participant list (GDPR compliant) will capture data around contact telephone numbers/ medical conditions.	**3X1 Low Risk**

Slips, trips, falls	Participants	Physical injury due to falling	**4X4 High Risk**	Appropriate footwear will be worn. Any incidents will be reported via the incident report form. Each participant will carry an appropriate first aid kit with them.	**3X2 Low Risk**
Accomodation	Participants	Cancellation, loss of deposit, fire whilst there.	**4X3 Medium Risk**	Ensure all deposits/payments are made in a timely manner. Familiarise with fire exits, fire evacuation and assembly point.	**3X1 Low Risk**
Catering	Participants	Allergies/ dietary requirements/food poisoning	**4X4 High Risk**	When preparing homemade foods, we will adhere to the information around food hygiene safety available on the SU website.	**2X2 Low Risk**
Medical emergency	Participants	Illness, injury, hospitalisation	**4X4 High Risk**	We will acclimatize ourselves with the Durham SU Emergency procedures and in the event of an emergency we will follow these procedures and report the incident using an incident reporting form.	**2X2 Low Risk**
Incorrect handling of equipment	Participants, local wildlife	Injury, littering, incorrect disposal of equipment	**4x3 Medium Risk**	We will acclimatise ourselves with the steps involved in the collecting and analysing of the soil samples. Once finished we will ensure that nothing is left at the sites, and following analysing we will ensure that equipment is disposed of correctly as stated by the manufacturer	**2x2 Low Risk**

Handling of chemical reagents for active carbon test (potassium permanganate, calcium chloride)	Participants	Injury, damaging of equipment. Potassium permanganate may intensify fire, cause skin burns, eye damage and organ damage with prolonged exposure. Calcium chloride can cause eye irritation.	**3X9 Medium Risk**	Very small amounts of reagents are used in very low concentrations, under suitable supervision and in appropriate facilities. Gloves and safety goggles will be worn. Potassium permanagate will be kept away from sources of heat. Reagent solutions will be disposed of appropriately (down the sink with water).	**3X3 Low Risk**
Soil test kit handling	Participants	Skin and eye irritation	**2x2 Low Risk**	Safety instructions from the test kit will be followed. Safety goggles and gloves will be worn.	**1x1 Low Risk**
qPCR	Participants	Damage to equipment	**1x2 Low Risk**	qPCR will be conducted under appropriate supervision after training	**2X2 Low Risk**

Fieldwork	Participants	Participants becoming lost, injury.	2x2 LOW RISK	Participants will have the phone numbers of Living Labs staff in case of emergency and will be oriented to the field site by someone familiar with the area.	1X1 LOW RISK

Health and Safety Risk Matrix

			Probability/ likelihood of risk realisation				
			Almost Impossible (1)	Not Likely to occur (2)	Could occur (3)	Known to occur (4)	Common occurrence (5)
		Health and Safety	A freak combination of factors would be required for risk to be realised	A rare combination of factors would be required for risk to be realised	Could happen when additional factors are present otherwise unlikely to occur	Not certain to happen but an additional factor may result in risk being realised	Almost inevitable that risk will be realised
Potential Consequences	Severe (5)	One or more fatalities. Irreversible health problems	5	10	15	20	25
	Major (4)	Partial or medium-term, disabilities or major health problems	4	8	12	16	20
	Moderate (3)	Lost-time injuries or potential medium-term health problems	3	6	9	12	15
	Minor (2)	Minor, very short-term health concerns on recordable injury cases.	2	4	6	8	10
	Insignificant (1)	Inherently safe, unlikely to cause health problems or injuries	1	2	3	4	5

Extreme risk	High risk	Medium risk	Low risk

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